BGIN Block #10 Meeting Report

- BGIN is a multi-stakeholder discussion body / distributed think tank.

Layer 2 Governance: Keynote

bg in

blockchain
governance
initiative
network

The Global Network for Blockchain Stakeholders™

© 2023 BGIN (Blockchain Governance Initiative Network) is registered as BN Association, a Japanese general association with its principal place of business at Shiba Building, 704, 4-7-6, Shiba, Minato-ku, Tokyo. All rights reserved.

The know-how described in this document was made available from contributions from various sources, including members of the BGIN and others. Although the BGIN has taken steps to help ensure that the know how is available for distribution, it takes no position regarding the validity or scope of any intellectual property or other rights that might be claimed to pertain to the implementation or use of the know how described in this document or the extent to which any license under such rights might or might not be available; neither does it represent that it has made any independent effort to identify any such rights. BGIN and the contributors to this document make no (and hereby expressly disclaim any) warranties (express, implied, or otherwise), including implied warranties of merchantability, non-infringement, fitness for a particular purpose, or title, related to this document, and the entire risk as to implementing this document is assumed by the implementer. The BGIN Intellectual Property Rights policy requires contributors to offer a patent promise not to assert certain patent claims against other contributors and against implementers. BGIN invites any interested party to bring to its attention any copyrights, patents, patent applications, or other proprietary rights that may cover technology that may be required to practice this document.

Block 10 D1_3

Overview

Keynote Session Report: Layer Two Governance in Blockchain Technology

Date: Tuesday, 5th March 2024

Location: Tokyo

Duration: Approx. 35 minutes

Summary

Meeting Report on Layer 2 Governance Keynote Address

The keynote address focused on the intricate and evolving topic of Layer 2 (L2) governance within blockchain technology. The speaker explored how L2 governance mechanisms can be optimized to enhance the security, longevity, and robustness of decentralized systems, offering insights into the dual nature of governance as both a powerful tool and a potential attack vector.

Key Themes and Discussions

The Essence and Challenges of L2 Governance

- Governance as a Problem-Solving Tool: Governance in the context of L2 is portrayed as a versatile solution to numerous challenges, including bug fixes, feature additions, and incentive creation. It operates by modifying the rules and logic of smart contracts, an action that is facilitated either through governance multi-signature wallets (multisigs) or governance tokens.
- Governance as an Attack Vector: The address highlighted a crucial downside to governance—its potential to serve as an attack vector, introducing security vulnerabilities not present in Layer 1 (L1) protocols.

Minimization of Governance

- The speaker advocated for the minimization of governance in security-critical aspects of blockchain infrastructure to retain the majority of L1 security properties. The address pointed to Uniswap as an example of an application that achieves robust security without extensive governance.

Infrastructure of L2 and Governance Application

- Layered Infrastructure of L2: L2's infrastructure consists of three layers: data availability, sequencing, and execution, each of which offers different governance minimization opportunities.
- Opting for Internal Resources: By utilizing L1 resources for data availability, sequencing, and execution, L2 solutions can minimize the need for governance and bolster security through a unified security assumption.

Governance for Human-Centric Aspects

- The address delineated a crucial role for governance in bridging the on-chain machine world with the off-chain, human-centric world. This duality presents opportunities for governance to manage public goods funding and decentralized front ends, showcasing how governance can be strategically used to handle aspects that inherently require human oversight without compromising the core security of L2 protocols.

Questions and Further Discussions

- The Q&A session delved into technical considerations, such as the distinction between various L2 solutions (e.g., rollups and validiums) and the implications of governance on the longevity and trustlessness of these systems.
- Governance Synchronicity: A concept introduced to address the challenges faced by applications spanning multiple L1 protocols, highlighting the importance of alignment in governance mechanisms to ensure seamless operation and updates.
- Eigen Layer and Security Implications: Discussions also touched on the potential impact of new technologies like the Eigen layer on L1 and L2 governance, exploring how advancements could alter the security landscape and governance requirements of decentralized systems.

Concluding Remarks

The keynote offered a comprehensive overview of the current state and future prospects of L2 governance. By balancing the power of governance with the need for security and simplicity, the speaker outlined a vision for a more secure, efficient, and decentralized future. The insights provided serve as a valuable foundation for ongoing discussions and developments in the field of blockchain technology and governance.

The address concluded with a note of appreciation for the opportunity to share insights on L2 governance, inviting further questions and discussions to deepen the understanding of this complex topic.

Following the Keynote, there was a roundtable discussion furthering the topic exploration. See the below, links and transcript of this keynote address below.

Roundtable Document	Discourse Thread
BGIN Block #10 Meeting Report_Layer2 Governance_Roundtable	https://bgin.discourse.group/t/block-10-call-for-contribution-layer2-governance-day3/339/2

Transcript

00:02
Speaker 1
Firstly, thank you all for coming today to the IKP day of begin block ten. It's of course, amazing to be in Tokyo now. Thank you. So to begin with, we'll keep this fairly brief actually, as the opening address, as we've already running a little bit late, but the IKP working group, when I joined a year and a half ago, were working on the Solband token paper and there's been so much progress throughout the year and also prior to that. So I'd just like to quickly provide the purpose of the IKP working group in the opening address and then the scope of work that we'll be going through. And then finally I'll share the work that's been completed and what we're working on continually and we'll be focusing on today as well.

01:00
Speaker 1
So of course the purpose is to explore the approaches of digital identity access management using blockchain and distributed ledger technology, and then to build sort of a canon of online resources to support multistakeholder dialogue in building governance into this space, specific to the non financial applications of blockchain technology. Of course we're giving considerations to privacy preserving and traceability technology, as well as law enforcement resourcing. And this is to sort of fill the gap of common understanding with regards to blockchain and identity key access management. So of course the scope is we're working with not only decentralized permissionless blockchains, but also interested in exploring permission blockchains and more recently focusing on new token standards like the soulbound token standards, as well as zero knowledge proof technology. So in terms of the progress report, IKP has produced a number of documents.

02:05
Speaker 1
To begin with, we did the decentralized finance system and associated regulatory conditions considerations. Sorry, there was an incident response of decentralized custody, a case study report which was also completed in 2022, which is actually quite relatable to one of the sessions

today on incident response. There was an NFT study report part one, an introduction of use cases study report on ransomware, the study report on selective disclosure, which is also quite relevant to the work that we'll be discussing today, Soulbound tokens study report part one, building and embracing a new social identity layer, which was the work that brought me into IKP working group, actually. And then of course the proposal of principles of defi disclosure and regulation, token economy and compliance, and most recently the azure knowledge proof and its applications that a main discussant has been working on.

03:11
Speaker 1
So without further ado, I'd like to open the IKP working group day of begin block ten and thank you all for attending and participating. Of course, this is an open dialogue. Thank you very much. I would like to announce that the next keynote on the layer to gamma state is a very interesting, important topic at this moment. That keynote is provided by Justin Drake from Ethereum foundation. And so I will switch the screen to that Justin now. Okay.

04:25
Speaker 2
Hello.

04:25
Speaker 1
Good morning, Justin, can you share your screen?

04:28
Speaker 2
Yes, I'll just share my screen. Just a second. Share my screen.

04:37
Speaker 1
Up.

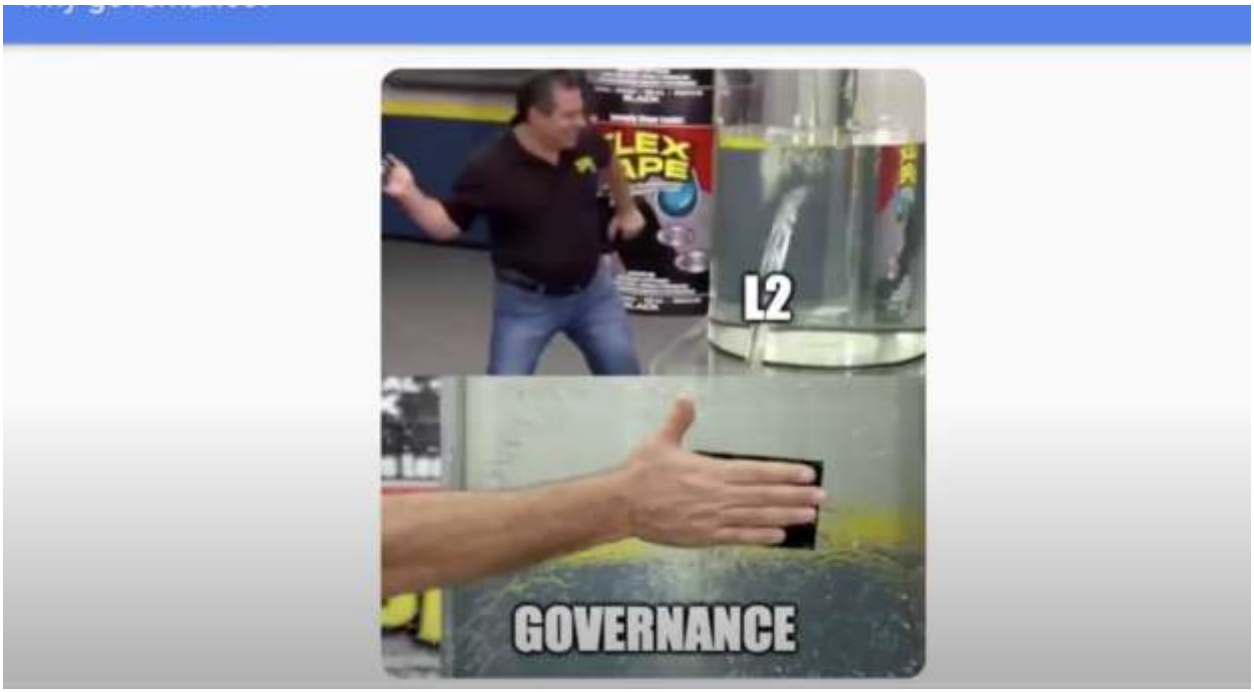
04:37
Speaker 2
There we go.

04:44
Speaker 1
It.

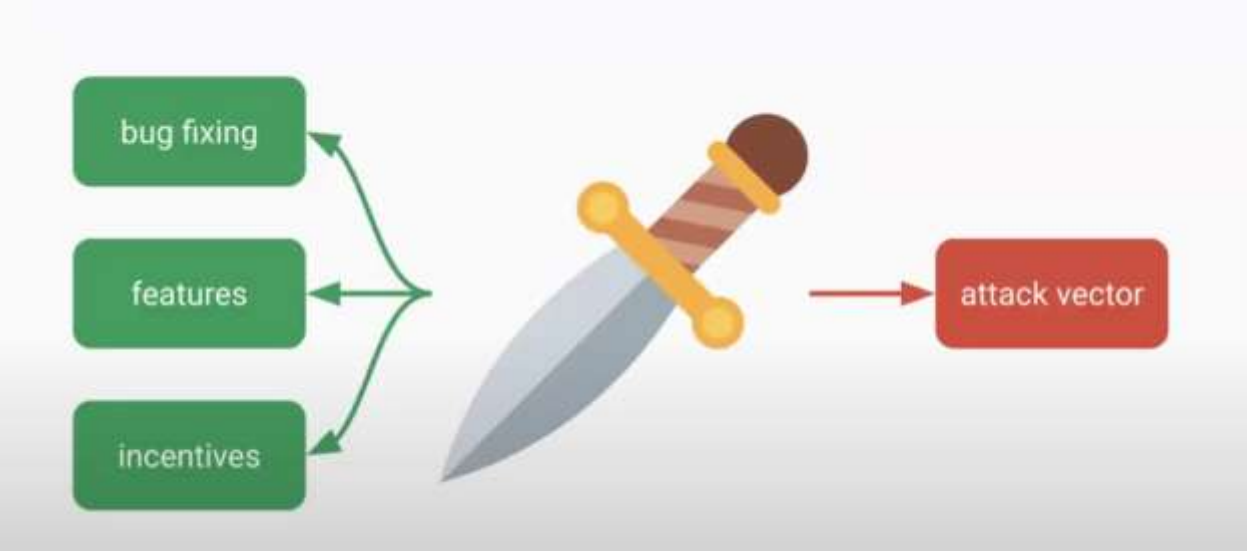
04:44
Speaker 2
Okay. Good morning. So I was asked to talk about layer two governance, and this is a topic I haven't talked about yet, so this is new for me, but I think it's an important and interesting topic. And what I want to try and communicate here is how I think about the harmony that we can try and achieve with l two governance. The governance was, if you will. So the talk will be in three parts. I'll cover and recap some of the basics of governance, and then I'll talk about what I mean by minimization of governance. Governance is a very powerful tool, but it comes with downsides. And I think it's important to minimize its use for security critical parts of the stack. And then the third part will be basically focusing on where I think governance shines the most.

05:43
Speaker 2
So why do we want governance? Why do we use governance? Basically from the perspective of an l two, we have all sorts of problems. And in some sense, governance is this magical tool that can solve any problem. You have a bug use governance, you need a new feature use governance, you need incentives use governance. And so in some sense, it's this kind of magical patch that can fix any l two governance problem. And just to make sure that we're clear on definitions, a governance system is all about changing the rules of the game. So changing the rules and the logic of a smart contract.

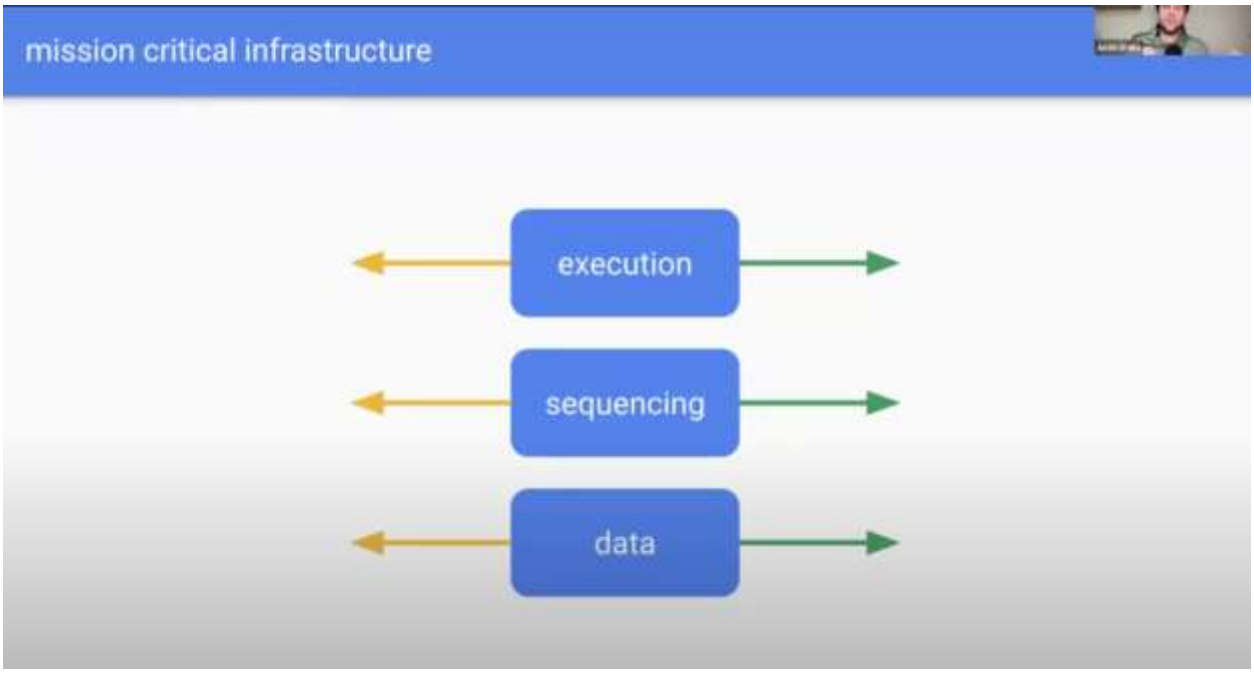
06:30
Speaker 2
And the way that we change the rules for l two s is either through a governance multisig, which is a set of semitrusted parties who can change the rules of the game, or governance tokens for more decentralized versions of multisigs, basically. And these two tools lead to change. Now, this very powerful ability to change the rules of the games, as I mentioned, is very powerful because it allows us to fix bugs and to add features and add incentives. But it also comes with one major downside, which is that governance is in and of itself an attack vector. And so if you want as an L two, to inherit the security of the l one, well, now you've introduced a new attack vector, which the L one doesn't have potentially, and that is a bad thing.



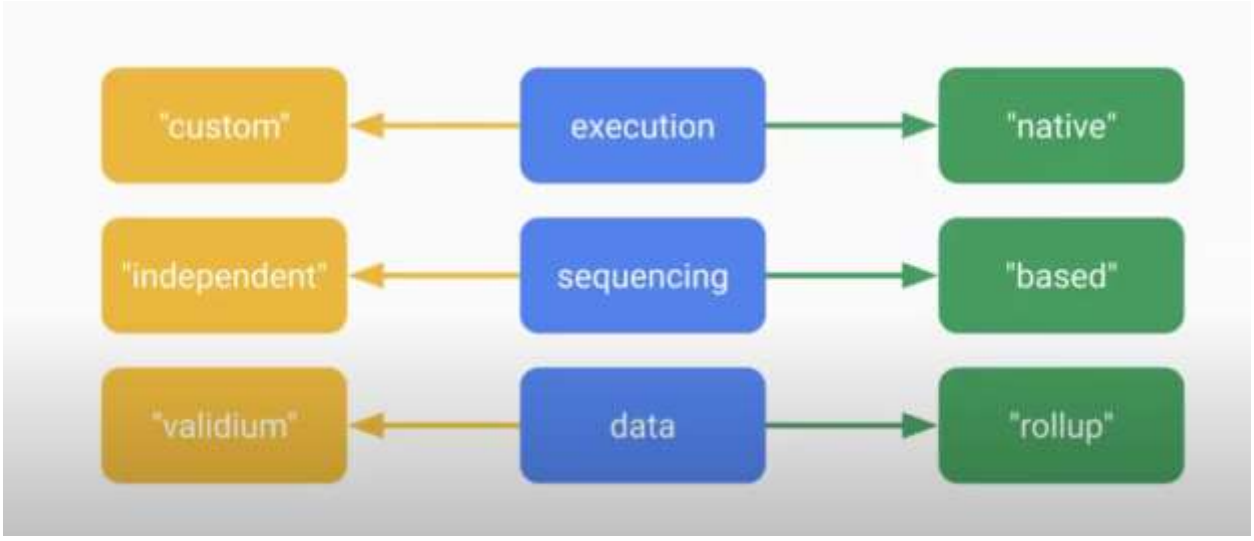
07:29
Speaker 2
And if you're interested in longevity, so if you're interested in building systems that will last for decades and centuries, that is a bad thing. If you're interested in maximizing credible neutrality and being maximally robust, that's also a bad thing. And so really I think governance is a tool that needs to be used very wisely. And we'll talk about when I think governance should be used and when maybe it should be minimized. So let me start with the minimization aspect here. And I think the whole point of l two s is to try and scale the L one and to retain the vast majority of the security properties of the L one.



08:12
Speaker 2
And I think one of the great examples of a governance minimized application, this is not an L two, this is a small contract, but I still think it's a useful example, is uniswap. So uniswap, you can think of it as being ethereum like in its security. It's an immutable contract. It inherits fully the security of the L one, and it provides this functionality of swapping without governance, or at least without governance for the L one contracts. Now let's look at an L two. What is an L two at a minimum. An L two for me is a piece of infrastructure with three layers. So there's a data availability layer. At the very bottom there's a sequencing layer which chooses the order of transactions, and then there's an execution layer which defines a virtual machine and an execution environment.



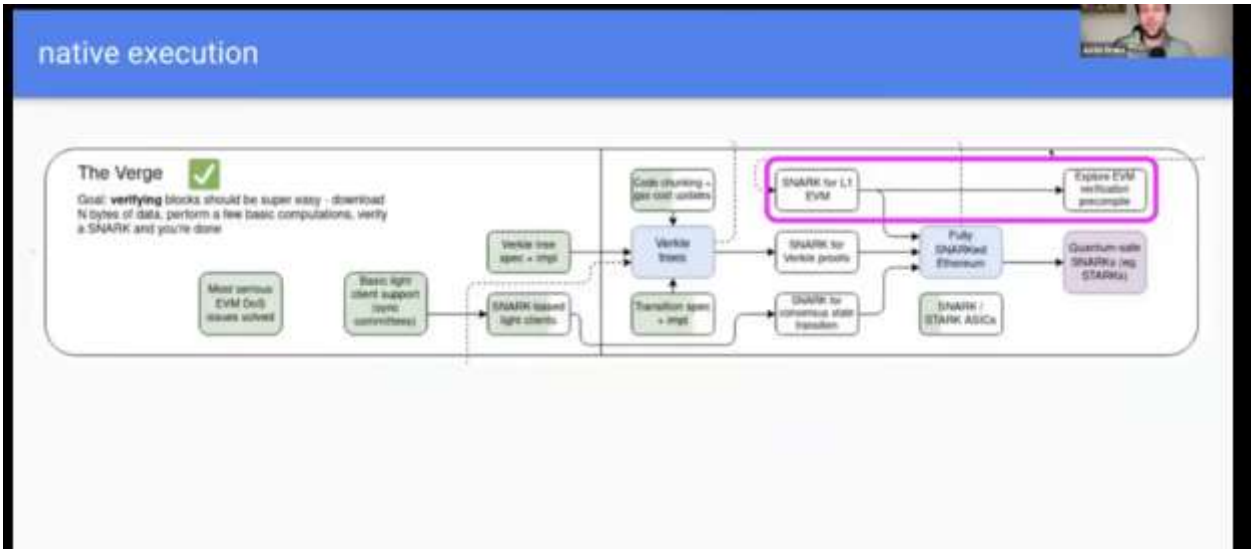
09:14
Speaker 2
And for each of these layers you have choices. So for example, for the data layer, you can choose to consume l one data, and you go on the right hand side, you become a roll up, or you can choose to not consume l one data, in which case you become a validium. And it's a very similar story with sequencing and execution. And we have all these different names that we give depending on which resources of the L one that you consume. So if you consume l one sequencing, we say that you are a based l two and similar for execution. If you consume l one execution, we'll say that you're a native l two. So in the case of Ethereum, you would be an EVM L two. And I'm going to try and describe a little bit why this has some benefits.



10:13
Speaker 2
So one of the reasons for going to the right hand side as opposed to the left hand side here is that when things break, you don't have to invoke governance. So for example, in the case of data availability. If you're consuming external data availability, which is different to Ethereum, there is a possibility that data availability layer will break down. And now you need to invoke governance to choose a new data availability layer. And so by choosing to opt into a roll up, you're consolidating security assumptions. You're basically using Ethereum both for settlement and for data. And so everything falls under one unified assumption. Another edge case that can happen is this external infrastructure, which is not ethereum kind of updating.

11:13
Speaker 2
So let's say that you're consuming, let's say Celestia as a data availability layer, and Celestia does a hard fork, which changes its rules of its game. And now potentially you have to invoke your governance as l two to match the change that has happened on Celestia. So one of the nice things about consuming internal resources is that you have an opportunity to minimize governance for these three critical pieces of the stack, data sequencing and execution. And if you consume all three of them, then I call them ultrasound roll up. So if you are a native based roll up, then you're an ultrasound roll up. Now what do I mean by consuming native execution? So if you look at Vitalik's roadmap diagram, there's this little section here, which is a little hidden around snarkifying the EVM.

12:14
Speaker 2
And one of the things that we can do is basically expose an opcode for l two s to verify EVM execution within the EVM. And this has a couple huge advantages. Advantage number one is that you can reuse a lot of the work of others as an l two, you don't have to re implement DVM and there won't be bugs by construction, right? You're consuming the EVM natively, and so there's no possibility for you to introduce a bug in your virtual machine if you want to be EVM equivalent. And then the other really nice thing is that if you want to be EVM equivalent, you don't need governance to keep track of Ethereum hard forks.



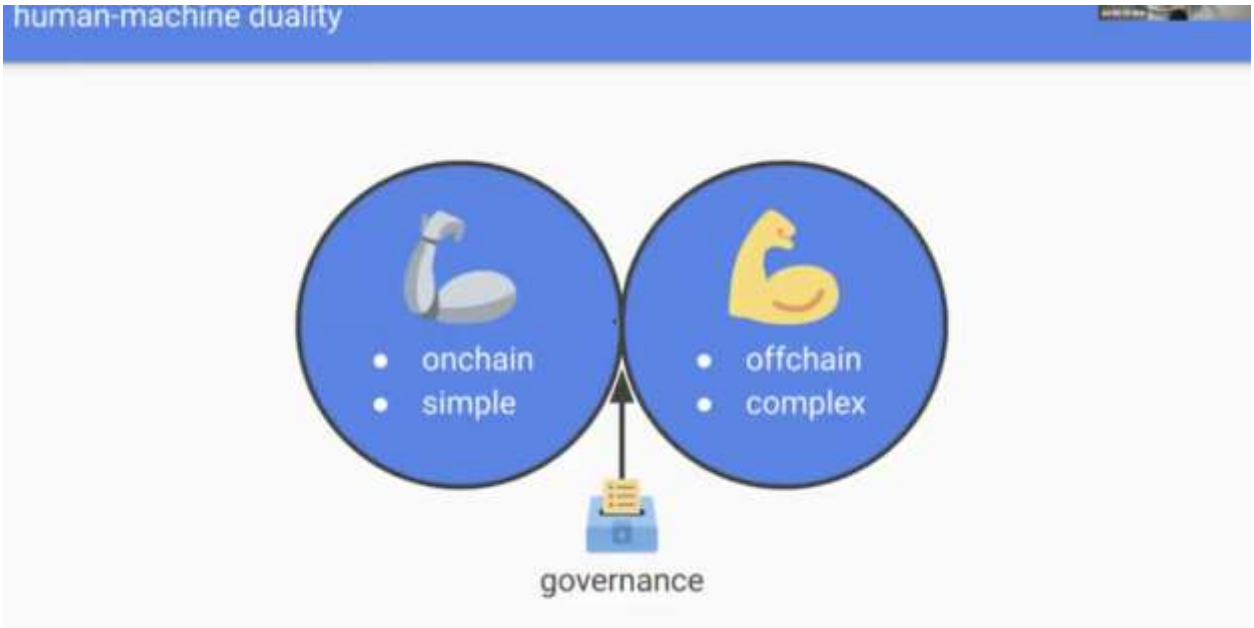
13:05
Speaker 2
So if Ethereum itself hard forks and changes the rules of the EVM, well now this pre compile will automatically update with the opcode, and so you don't need governance at all to manage your execution layer. And there's a couple of tools that we use in order to provide this pre compile for l two s. One is client diversity, which is a way to hedge against bugs. So if one client has a bug, that's fine because we still can rely on the other client and we can also make use of formal verification. So these are two tools that basically help us address the need for governance, specifically to deal with bugs. Now, the other one that I briefly mentioned is around sequencing. And this is a bit of a new idea that L two s can consume sequencing services from the L one itself.

14:15
Speaker 2

So right now, l two s mostly have centralized sequencers, but it turns out that they can use the l one proposers that are already sequencing ethereum transactions at L one. And one of the really nice things about this, but this is not the topic of this talk, is that we can have the various l two s enjoy synchronous composability among themselves. So it's a way to fix the fragmentation that we have on Ethereum today, where each l two is its own silo and we have fragmentation of liquidity and fragmentation of applications and users. Okay? So we've understood that governance basically is a very powerful tool. It means that you can basically change the rules of the game arbitrarily.

15:09
Speaker 2
And this is something that you want to avoid for the core logic that touches the security elements of an L two, which is the data availability, the sequencing and the virtual machine and execution. But I believe that there is a place for governance to thrive and to shine. And this is what the third part is all about. And really what we want to try and do is find a way to leverage the huge power of governance while at the same time being mindful of safety. And the way that I think about it in terms of the appropriateness of governance, is that we have this human machine duality, if you will. On the left hand side you have the machine world, the smart contract on chain world, which needs to be as robust, as simple, as secure as possible.

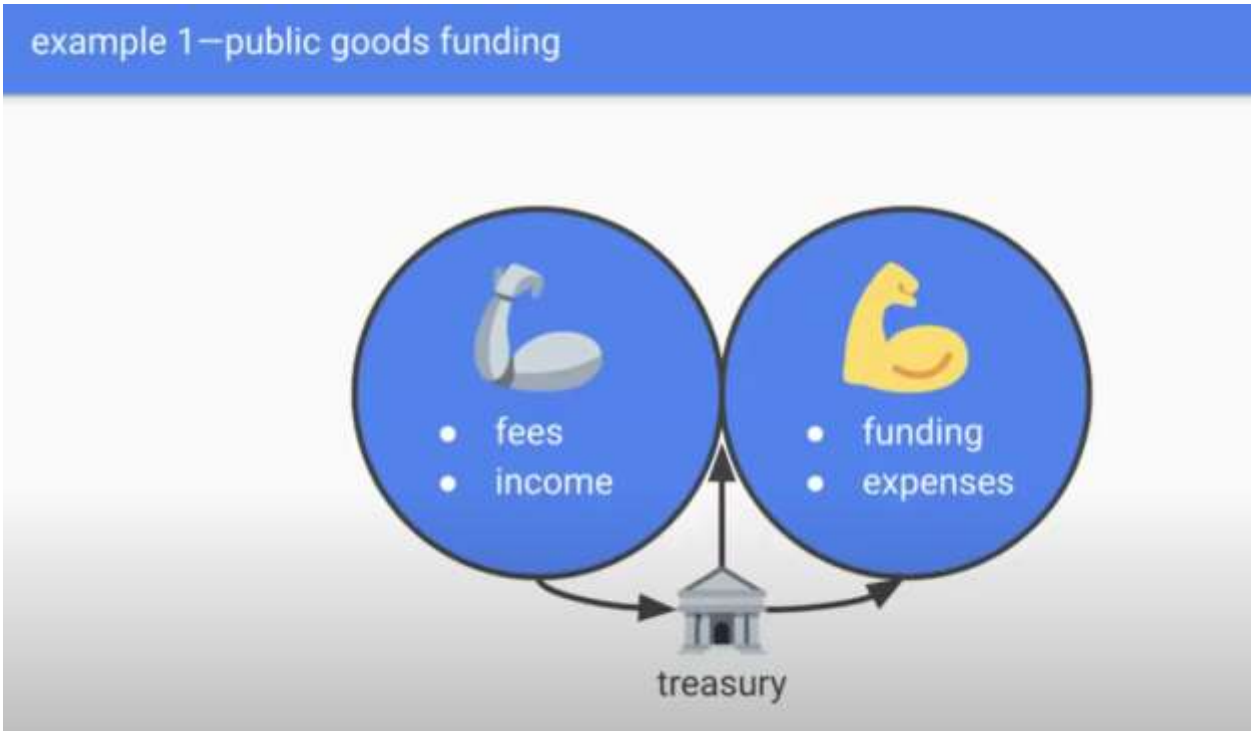
16:12
Speaker 2
And then on the other side, we have the messy, off chain, complex human world, where stuff also needs to happen. But it's kind of ogonal and disjoint world. And really the place that I think governance plays a key role is being this linchpin, being this intersection point between these two worlds, kind of bridging in a very safe way, these two worlds. And I'm going to give two examples. Example number one is around public goods funding. So imagine that we have a treasury within the layer two, which is funded, for example, through execution fees and income. So we have the machine world, which is fully autonomous, maximally secure, which generate these fees and is funding the treasury. And then on the other hand, we have a mechanism which is basically governance, which decides how to spend funds from the treasury.



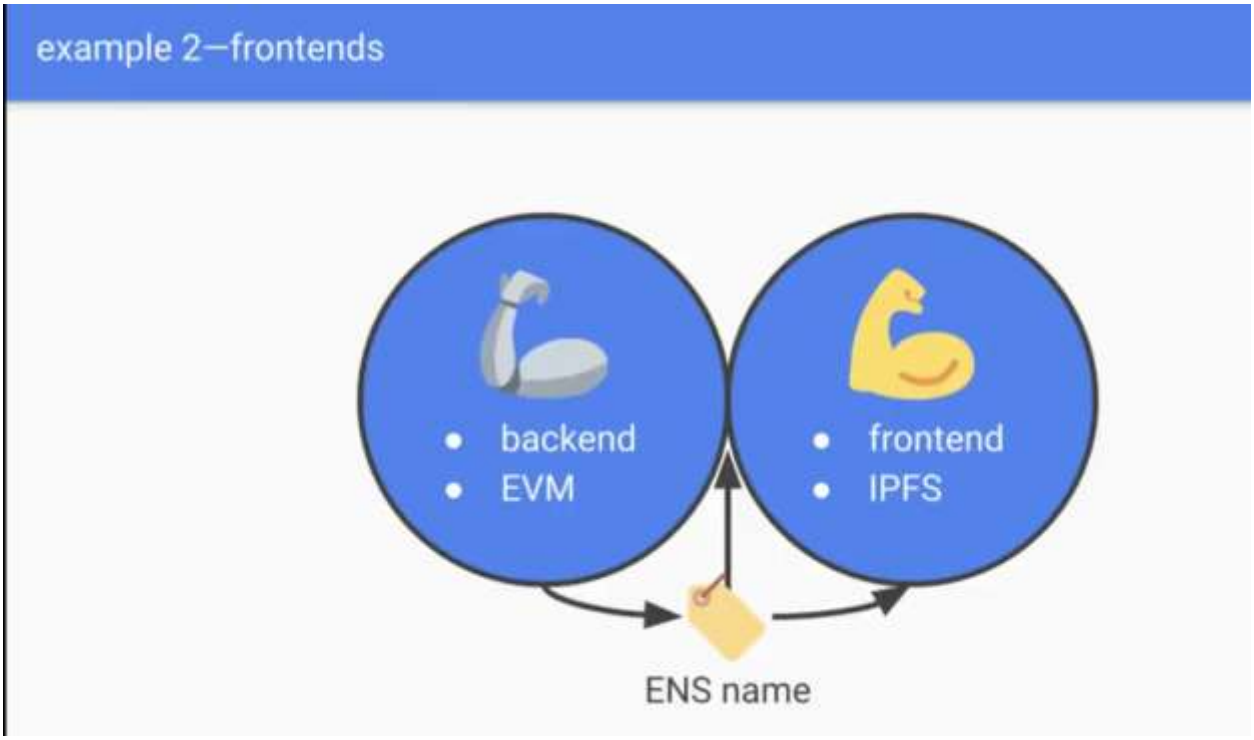
17:30

Speaker 2

And these funds, for example, can be spent towards public goods funding. And now notice that even though this treasury could get hacked, so for example, an attacker could try and 51% the governance token and try and steal all the funds in the treasury. In the worst case, the treasury gets emptied, but it doesn't destroy the l two. All the security properties of the l two remain, all the smart contracts remain. It's just this graceful degradation whereby the treasury functionality no longer works. And so sure, maybe future public goods funding is compromised, but at least the core guarantees and properties of the l two are not compromised. So that's example number one, where we have this minimal piece of infrastructure on chain, which is the treasury, which is governed through a governance token. And then the second, I think, interesting example, is around front ends.



18:42
Speaker 2
So right now we have a bit of an awkward situation where the back end think of Uniswap. The uniswap smart contracts are maximally robust and secure, but the front ends are controlled by uniswap labs. And that doesn't have to be. We can also have decentralized front ends. And one way to do that is basically to have an asset, namely an ens name. So for example, uniswap ETH, which is controlled by governance and which points to a front end. So that could be, for example, an ipfs hash. So front ends need to be updated by humans, just like treasuries need to be maintained by humans. But that's fine. We can have humans basically make proposals for changes to the websites. And the way that they submit governance proposals is by basically providing ipfs hashes for new versions of the front end and the website.



19:49
Speaker 2
And if the governance that controls the ens name is happy with the change to the front end, it can vote it in. And then, ta da, the website, which you access just by typing in uniswap ETH in your browser, automatically loads the new version of the website. And so here we have this decentralized and governance controlled front end, which is really cool because now we have a fully decentralized application, as opposed to just the back end being decentralized. So that's it. I only have 20 minutes. Happy to answer some questions, and I hope that I've tried to basically distinguish two use cases of governance.

20:36
Speaker 2
One, which is a bit of a short term necessary evil, where sure, in the next few months and years we will need governance to be able to fix bugs that are core to the L two s and touch, and that are security critical. But over time, we can remove governance from that core aspect of L two s and instead have governance focus on managing humans and the messiness of humans, either through treasuries or in the context of decentralized front ends. Thank you.

21:14
Speaker 1
Thank you very much for your great keynote speech. So we have time for a couple of questions. Is there anyone who has a question to ask? Thank you very much for the talk. Is there something

like a custom independent validium solution? Or would that basically have then nothing to do with layer one at all?

21:50
Speaker 2
Right, so as you said, you can choose to be in a validium, you can choose to be independent. And every time you move away from the l one, then there's more and more of a need for governance in case things break. One of the concepts that I like is this idea of governance synchronicity. So each l one has its own governance. And so if your application straddles two l ones, for example, celestial and ethereum, then now you kind of need to manage this complexity. Whereas if you have everything within one l one, when that l one hard forks, you have a very clean situation where the application has to do nothing and it will just keep on working. Now, some applications, just through economic necessity, won't be able to afford to be a roll up. They'll have to be a validium, for example.

22:58
Speaker 2
And in this case, there's several possibilities. Possibility number one is there's a bit of a trade off space. Possibility number one is that you accept that you don't have longevity. So your validium is maybe perfectly trustless, but it can break at some point in the future. So if you're building a game, for example, it's maybe fine that the game only lasts a few years and then it breaks. And then the option number two is you do want this continuity, but now you have to bring in a governance mechanism to ensure that when the data validity layer changes or breaks, or whatever it is, you can have continuity. So you lose a bit of the trustlessness, but you gain in terms of longevity. Yeah, I hope that answers the question.

24:07
Speaker 1
Thank you very much. Is there any other question? Hey Justin, can you hear me? Yes. I enjoyed listening to some of your talks which were peripheral to these issues in Istanbul on the effect of Eigen layer and security validation, how the introduction of Eigen layer this year may change the security properties of Ethereum and Ethereum governance and may potentially have an impact on l two s now and future things that look like l two s in the future. I'd love to hear a few of your thoughts on how some of those ideas fit with what you've just shared this morning.

24:57
Speaker 2

Right? I mean, really, the L one and l two governances are very different. So l one governance is not token based, whereas on L two in some sense, the only way to do decentralized governance is to use tokens on the L one. Instead, we have layer zero governance, where human beings will just start running different nodes. So they're very different and in some sense orthogonal forms of governance. Having said that, there are exceptions. And basically the exception is that sometimes the l two s can pollute the L one because there's some sort of systemically relevant catastrophe that has happened. And I think the best example here is the Dow. You have an application which is so large in some sense is too big to fail, and that mandates some sort of bailout at the social layer.

26:07
Speaker 2
And what Eigen layer is offering is a huge amount of power and flexibility and programmability. But it does mean that we might see applications at layer two just consume a lot of collateral. And what that means is that there could be mass slashing events. Imagine 20 million e just suddenly destroyed just because of some sort of a bug or some sort of an attack. And I think if that were to happen, there is definitely a possibility that it's such a big event that it compels the l one governance to step in and do some sort of a bailout. Now, one thought that I have here is this idea of penalties capping. So with staking today, you can lose all your ETH, potentially in the worst case, in one fell swoop.

27:13
Speaker 2
And the proposal of penalties capping is to cap the penalties to one eight, for example, of your stake. And what that means is that in the worst case, only one eight of the roughly 31 million ETH which is currently staking would get slashed. And that's actually a reasonably manageable amount. It's 12.5%. And this amount is small enough that, more likely than not, the governance layer wouldn't kick in. Now, unfortunately, Eigen layer, it's completely permissionless. And so they can have whatever slashing rules they want. And so even though we could do penalties capping for staking, it doesn't mean that Eigen layer will do penalties capping for restaking.

28:00
Speaker 2
But maybe there's an opportunity here to have some sort of a social norm whereby we encourage restaking AVss to have this capping logic to limit some of the systemic risk and possibilities for the L zero to intervene.

28:26
Speaker 1
Is that good? Okay, another question. Yeah. Justin, thank you so much for this talk. I've learnt so much. Part of my question today is actually, with regards to some of the, of course, blog posts that went towards this governance minimalization. And I guess, where do you see the more important minimalization in L two as a priority? So if you were to say, for example, I'm launching a new l two and I want to prioritize governance minimalization, where do I rely on l one as the priority? And almost like a question increments, is that possible?

29:18
Speaker 2
Right. I think right now the most pressing is actually the execution layer. So you have l two s, for example, that don't have any fraud proof or any kind of proving whatsoever. So optimism is a great example. So here there's a huge opportunity to really step up and improve the trustlessness of the execution layer. One of the difficulties with the execution layer is that it will take a long time before we get this opcode, maybe half a decade. And the reason is that in order to get this really high client diversity for ZKvMs, and by the way, that's type one Zkvms, and in order to do all this formal verification work, it's going to take a very long time.

30:14
Speaker 2
So one of the things that the L two s have started doing is having a little bit of this client diversity through multi proving, which is a great first step. So for example, scroll is now a dual prover roll up. It has a ZK pot which verifies the snarks, and it's possible that has bugs, right? Because the ZK circuits are extremely complex, the proof systems are extremely complex. And so it's more likely than not that there are critical vulnerabilities in that layer of the stack. And then there's another counterpart to the prover, which is an SGX prover. So here you're using trusted hardware, SGX, and it's much less likely that there are bugs there because all you have to do is cross compile GEF and run it on SGX in order to verify these state routes.

31:12
Speaker 2
And so now as an attacker, if I want to go and attack the execution layer of scroll, I have to not only find a critical vulnerability in the snarks, but also break SGX. And these are like completely different problems, both of which are fairly hard. And being able to solve both as an attacker is extremely difficult. Now, in terms of client diversity, eventually what we want is every single execution client. So Gef, ref, Besu, and all the other execution clients to each have their own

implementation of Zkevm, and then having this robustness through clients kind of fixing themselves without a hard fork, just like whenever there's a bug in one of the clients, the developers push a fix, and it's not critical because it's one of the minority clients.

32:18
Speaker 1
Thank you. Has a question. I may ask one more question. Just to have your support in teeing up the conversation that we're about to have on this topic, where do you think we should explore in terms of sort of debating the issue of governance on L two s? I'm not sure if you need a bit more context around what begin does, but we're trying to build multistakeholder dialogue. There's a lot of different participants in the room and diversity of ideas. So while this is a little bit more technical on the l two l one dynamics of governance, do you think there's something that we can support, almost like support you after our debate?

33:13
Speaker 2
Yes, I think the thing where there's the most lowest hanging fruit, and I'm a little surprised that it hasn't happened yet, is this idea here. So I think it would be really cool if we had these decentralized front ends. And all that needs to happen really, is for the governance layer to have access to EnS names and for there to be some sort of a standard to point EnS names to ipfs hashes. And then once you have this standard infrastructure, then you have decentralized front end. Now, in terms of the multi stakeholder aspect of coordination, well, we need browsers to opt into this. So we already have browsers like brave that have integrations with ENS, but it would be great to have more of the browsers play in.

34:16
Speaker 2
We obviously need the big d five projects and l two s to opt into this kind of standard where the ens name is controlled by governance. And then I think there's the developers of IPFs and protocol labs that are very keen. Actually, they messaged me a few days ago around exactly this idea where you have decentralized front ends, and if you want to be put in touch with them, I'd be more than happy to introduce you to the developers behind IPFs and Filecoin.

34:59
Speaker 1
Thank you so much. Okay, so there's another question from remote participant. So for remote

participants, if you have any question, please raise your hand. No. Let's thank to Justin for your great invited talk.

35:29
Speaker 2
Thank you so much. Thanks for having me.

35:30
Speaker 1
And so we have the following workshop session on the layer two governance. I think that for Justin in your time that it's a midnight, you don't have to remain this room. So we will continue our discussion in this room. Anyway, thank you Justine.

35:50
Speaker 2
Thank you.

For access to the Keynote Speech see:

[LINK TO RESOURCES IN DISCOURSE]